

JU SeqWorkbench Alpha

User Guide

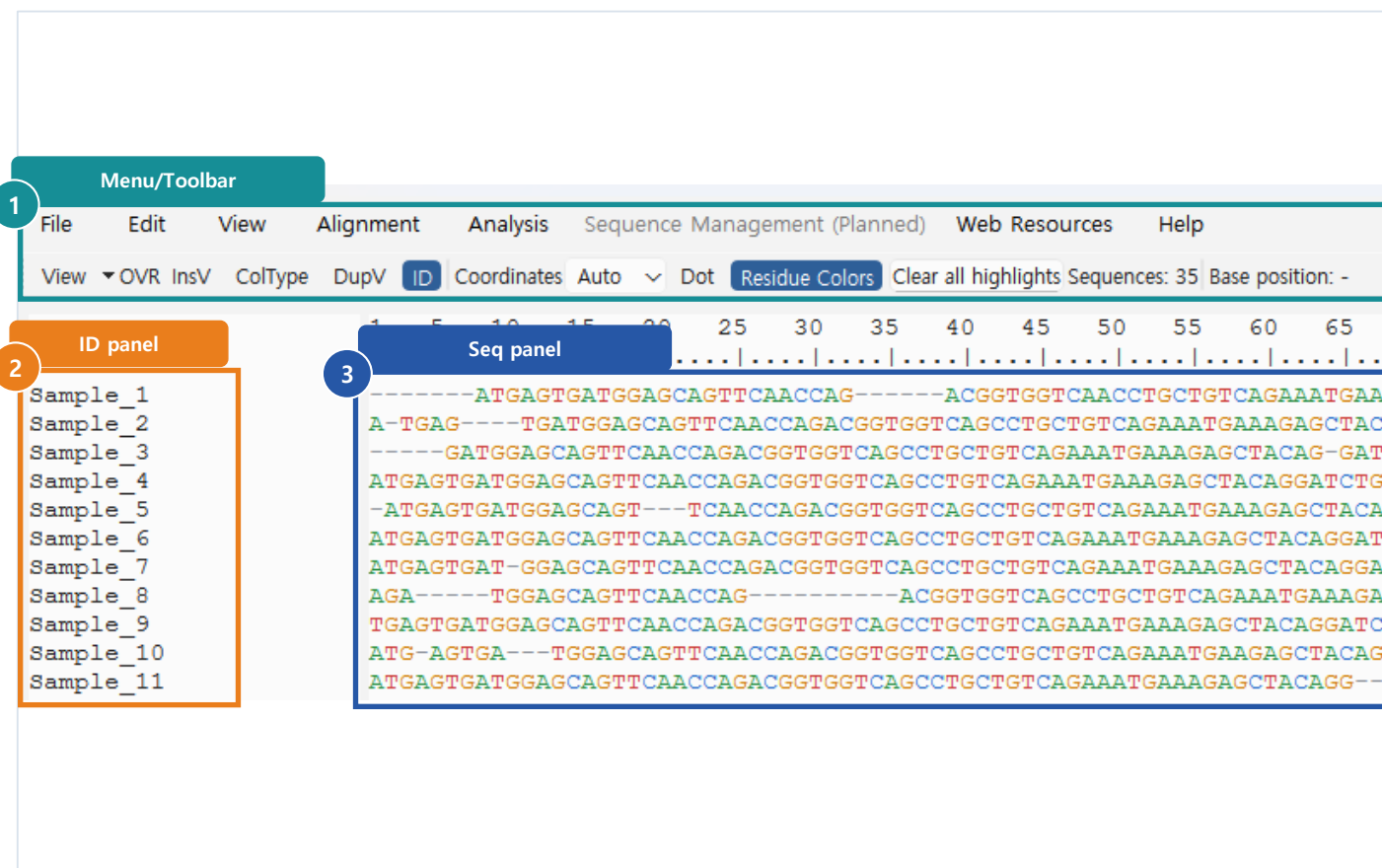
A local desktop workflow for reviewing and editing Sanger/FASTA/MSA sequences, then inspecting mutation patterns at selected positions and regions as tables and figures.

Guide Purpose

This guide focuses on key screens and interactions so first-time alpha testers can quickly understand what happens when they click each area.



Main Viewer Layout



Core Areas

Menu: file, view, analysis, and help functions

Tool bar: edit state and quick actions

ID panel: strain / sequence ID selection

Seq panel: nucleotide or amino-acid sequence display

Key Toggles

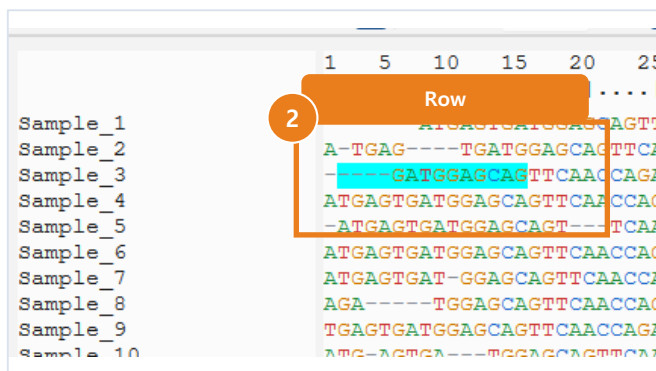
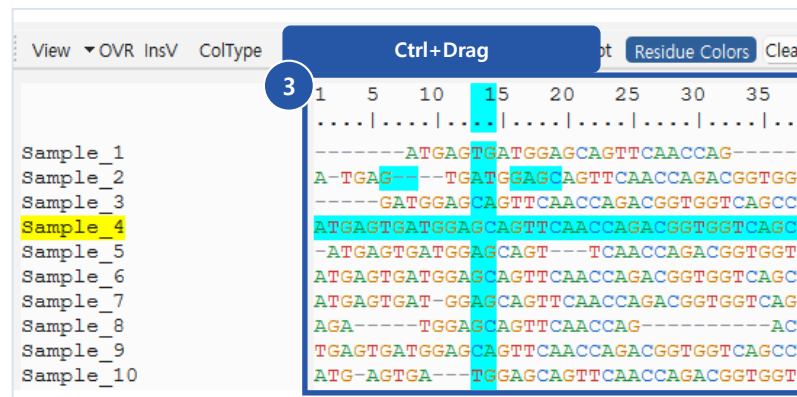
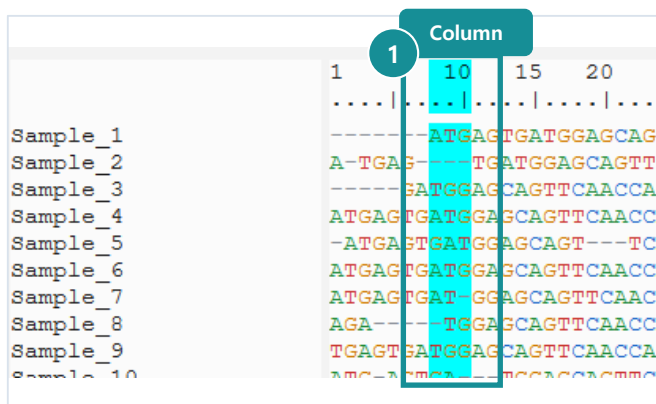
View/Edit, INS/OVR, Dot, Residue Colors, Coordinates, ID panel visibility, and Clear all highlights directly affect screen behavior and editing targets.

DupV: Duplicate the current view window

ColType: Batch input and deletion based on cursor or column

InsV: Paste based on cursor or column

Highlight and Selection Interactions



Alpha Limitations

Some drag selections may extend beyond the intended strain/row boundary. When multiple highlights exist, paste targets the most recently active highlight.

Basic Behavior

Drag sequence area: horizontal highlight

Drag ruler: column highlight

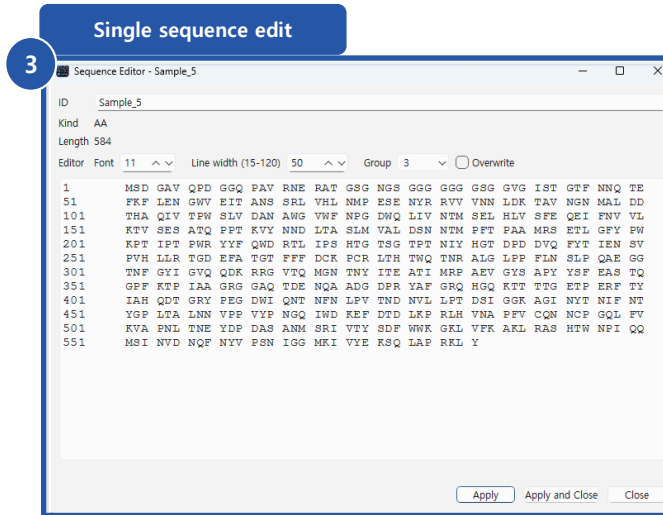
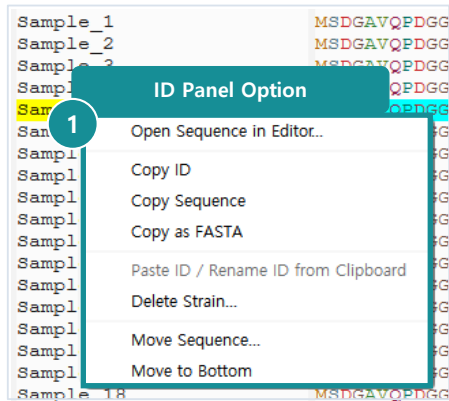
Ctrl+Drag: accumulated highlight

Clear all highlights: remove all highlights

Backspace/Delete: Deletes the currently active editable BODY or COLUMN highlight target.

Ctrl+Backspace/Delete: Deletes all explicitly selected editable BODY/COLUMN highlights in one operation. ID-panel highlights are never included in sequence deletion.

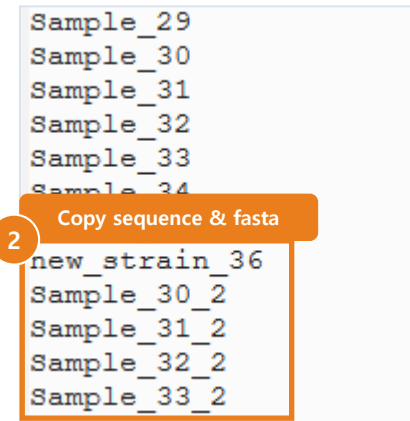
ID Panel Interactions



Basic Behavior

Open a dedicated editor for a single sequence, or copy and paste its ID and sequence data.

To reorder a sequence, select the move option and then click a different sequence ID. The selected sequence will be moved directly above the sequence you clicked.



Single-Sequence Editing

Edit the sequence ID or modify the nucleotide sequence directly.

Display options for nucleotide data can be configured in the **Editor** settings.

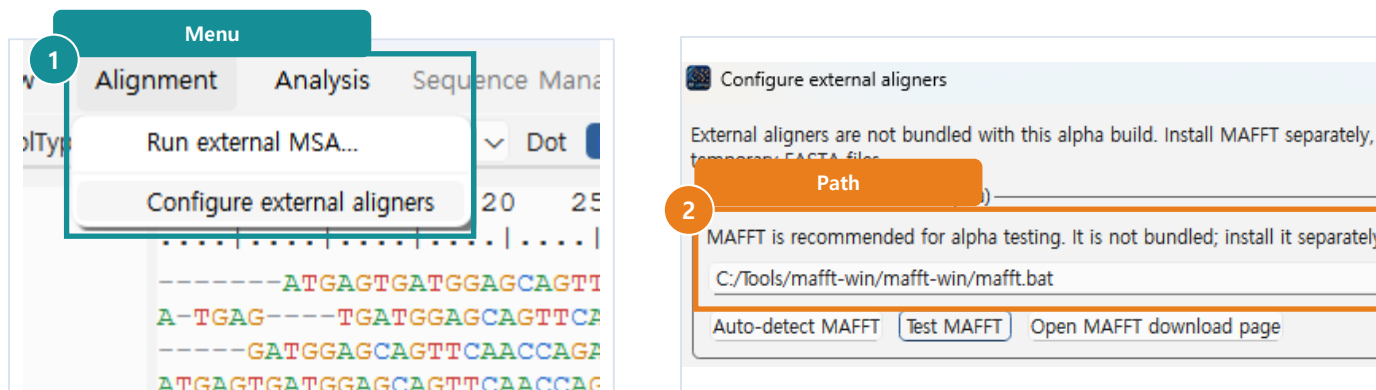
Copying and Pasting Sequences

Copy ID: Copies the sequence ID.

Copy Sequence: Copies only the nucleotide sequence data. (*Single sequence only.*)

Copy as FASTA: Copies both the sequence ID and nucleotide data in FASTA format.

External MSA Tool Setup



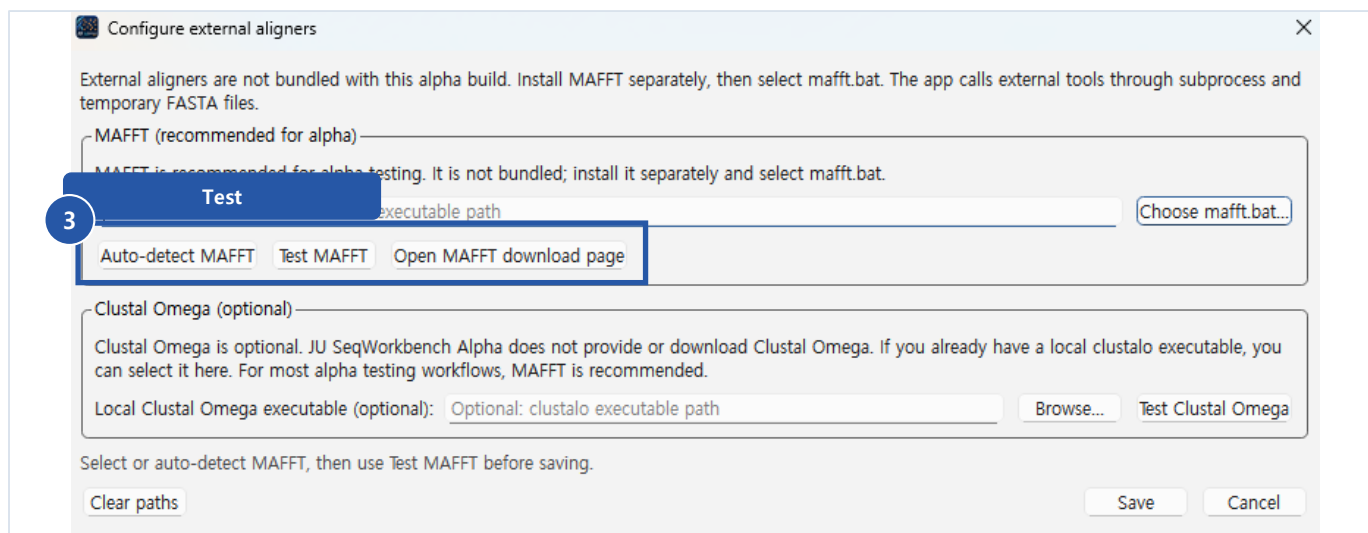
Setup Flow

Alignment > Configure external aligners

Select the MAFFT executable path

Use Test to confirm the executable works

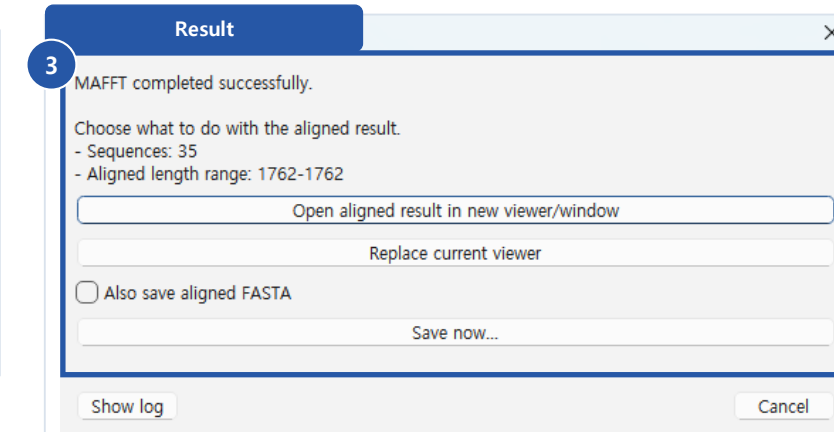
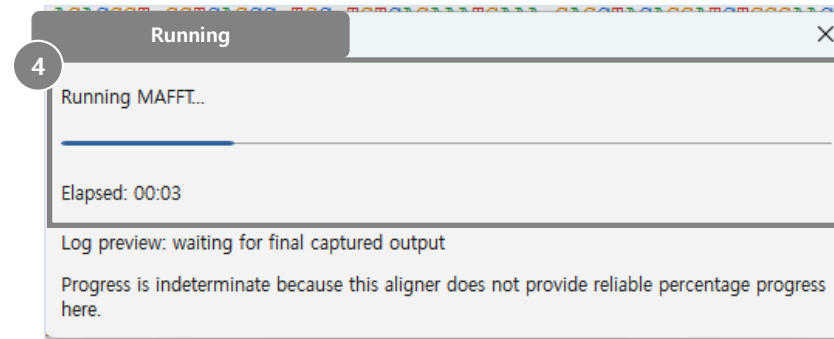
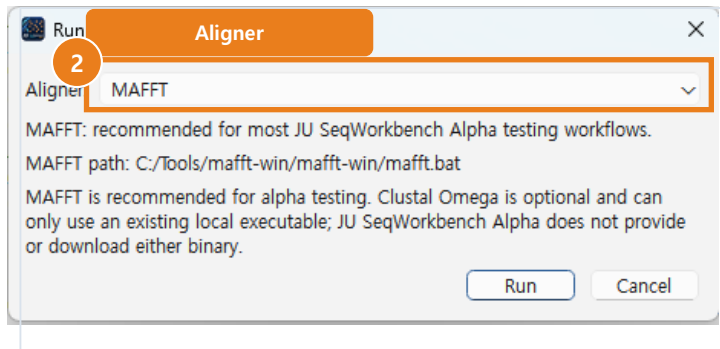
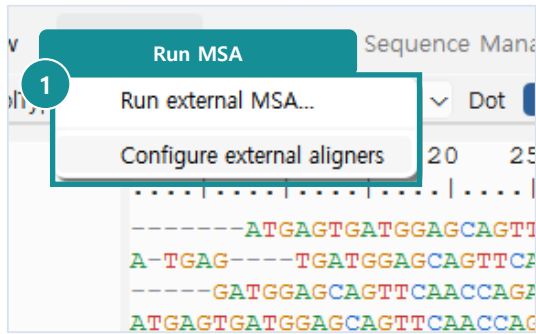
*The path is saved as a local user setting



Distribution Policy

The alpha distribution does not include MAFFT. Users connect separately installed local executables. Clustal Omega is currently disabled in Alpha.

Running MSA and Applying Results



Note

Project files (.dvproj) are not aligned directly. The app creates a temporary FASTA from the current viewer working sequences and passes it to the external tool.

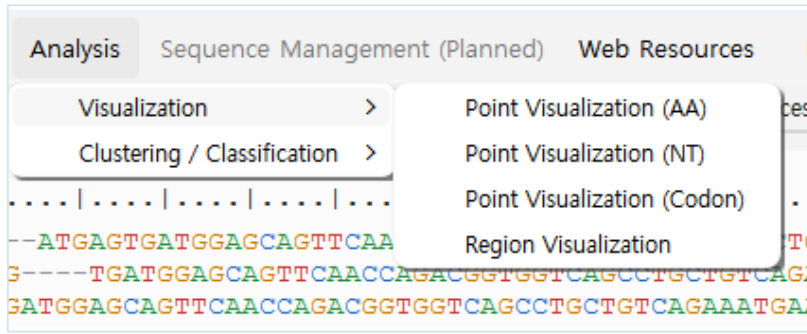
Running

By default, MSA uses all working sequences. When two or more strains are selected, only the selected strains are aligned. A single selected strain cannot be aligned. Mixed NT/AA input is not allowed. For a subset alignment, the result can be opened in a new viewer, but Replace Current is unavailable.

Result Handling

After running MSA, review the output in a separate result window. You can choose whether to replace the current viewer or open the result in a new viewer.

Point Visualization: Setup

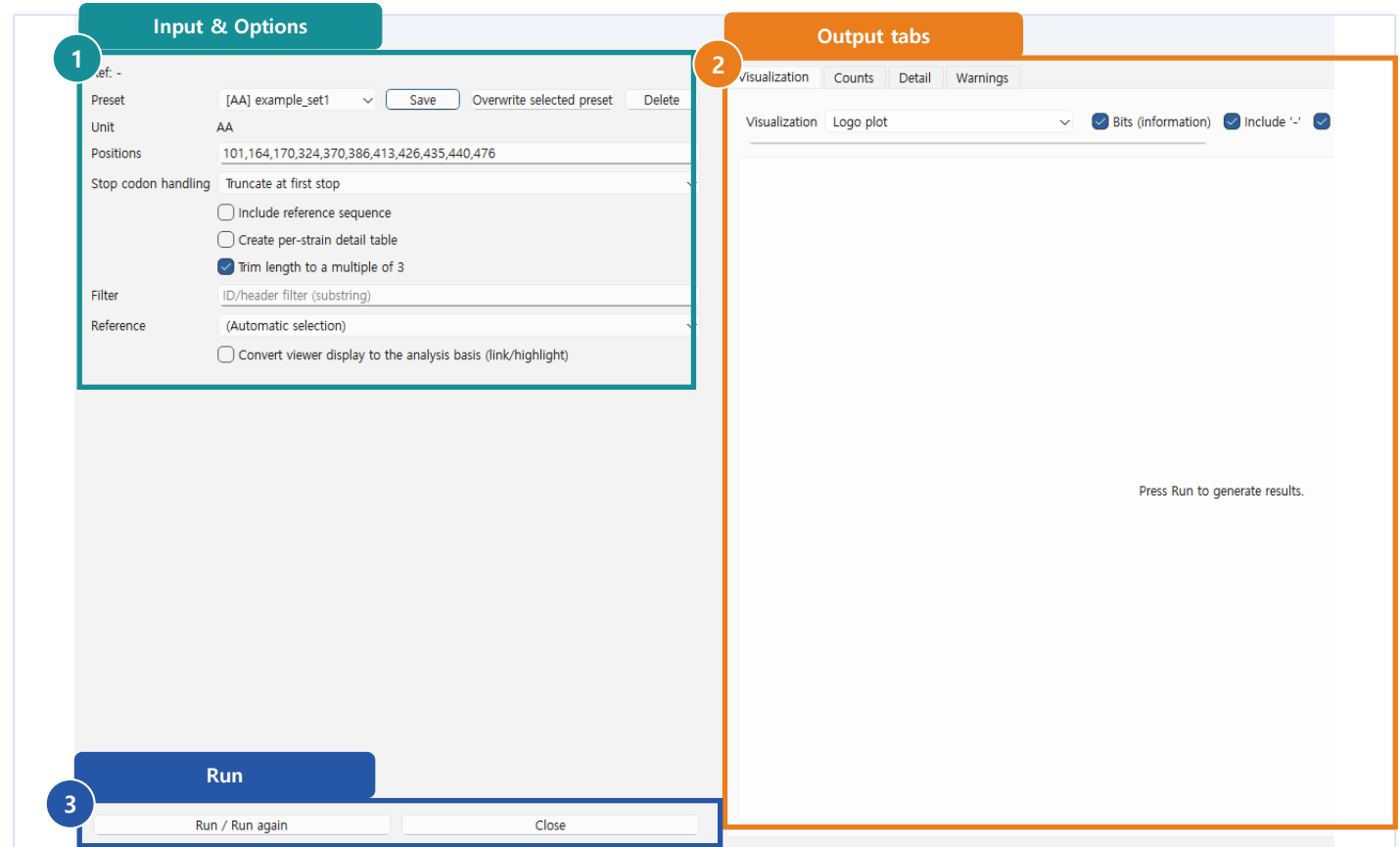


Input Panel

Set Preset, Unit, Position, Stop codon handling, Filter, and Reference. Coordinate meaning changes depending on AA, NT, or Codon mode.

Output Panel

For the same set of positions, switch among logo plots, mutation maps, entropy, major allele frequency, composition bars, and other output types.



Point Visualization: Figure Adjustment

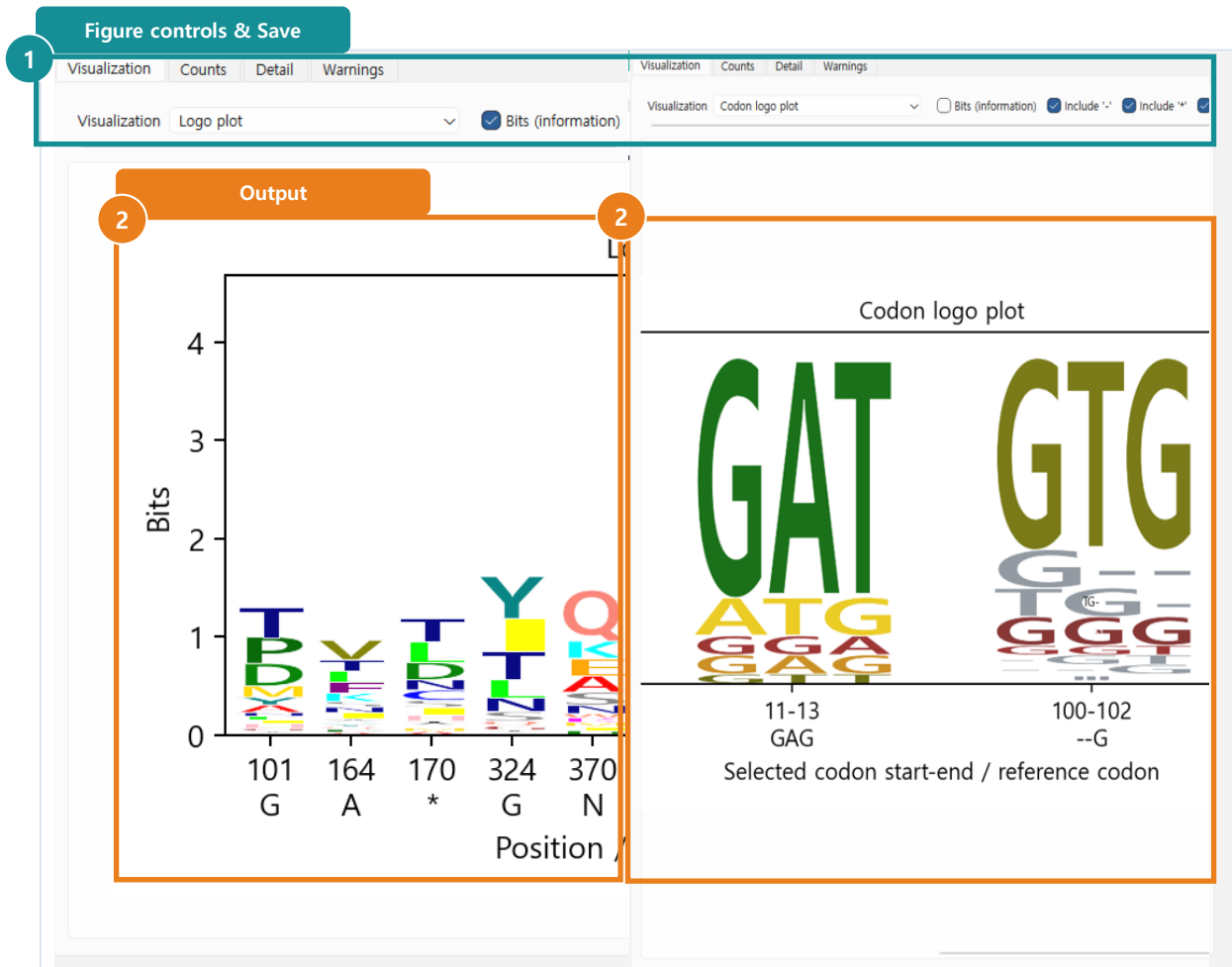


Figure Options

Adjust the title, axis labels, ticks, scale, text size, font, and style, then redraw using Rerender.

Visualization area

Multiple visualization modes are available, including Logo Plot. The rendering style may vary depending on the selected visualization mode and sequence unit (NT, AA, or codon). For example, a codon logo plot displays each codon as a three-character token.

Save

Use Save figure to export PNG / JPG / PDF / SVG images. For papers or reports, add numbering and captions later in PowerPoint or other layout tools.

Point Visualization: Tables and Viewer Highlighting

1 Result table

View Group by variant (IDs in one cell) Filter Press Ctrl+F to focus; type to filter Clear Sort: Input order

	N	Position	RefAA	MutAA	SeqIDs
0		101	G	*	
1		101	G	A	
2		101	G	D	Sample_10;Sample_14;Sample_23;Sample_24;Sample_35;Sample_8
3		101	G	H	Sample_4
4		101	G	I	Sample_27
5		101	G	L	Sample_5
6		101	G	M	Sample_11;Sample_2;Sample_20
7		101	G	N	Sample_18

3 Position

2 Viewer highlight

	75	80	85	90	95	100	105	110	115	120
Sample_1	TCTFKYARK*KL*KSGCK*YG*NCS*RKHGFR*YSC	TNCNTLVIG*CKCLG								
Sample_2	I*ICQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_3	NMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_4	MPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_5	VHLNMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGV								
Sample_6	NMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWS								
Sample_7	KYARK*KL*KSGCK*FG*NCS*RKHGFR*YPC	TNCNTLVIG*CKCLGSLV								
Sample_8	HLNMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGV								
Sample_9	ICQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_10	LNMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGV								
Sample_11	CQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_12	ICQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_13	YI*ICQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_14	HLNMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGV								
Sample_15	NMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_16	NMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_17	RK*KL*KSGCK*FG*NCS*RKHGFR*YPC	TNCNTLVIG*CKCLGSLV*SR								
Sample_18	K*KL*KSSCK*FG*NYS*RKHGFR*YPC	TNCNTLVIG*CKCLGSLV*SR								
Sample_19	YARK*KL*KSGCK*FG*NCS*RKHGFR*YPC	TNCNTLVIG*CKCLGSLV*SR								
Sample_20	CQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_21	ICQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_22	NMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_23	HLNMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGV								
Sample_24	LNMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGV								
Sample_25	YARK*KL*KSGCK*FG*NCS*RKHGFR*YPC	TNCNTLVIG*CKCLGSLV*SR								
Sample_26	PESENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_27	*ICQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_28	ICQKVKIIEEWL*IIWIKLQLETWL*MIPMHEL*HLGH	WLMQMLGEFGLI								
Sample_29	NMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_30	NMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_31	PESENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGVWF								
Sample_32	TCTFKYARK*KL*KSGCK*FG*NCS*RKHGFR*YPC	TNCNTLVIG*CKCLG								
Sample_33	YARK*KL*KSGCK*FG*NCS*RKHGFR*YPC	TNCNTLVIG*CKCLGSLV*SR								
Sample_34	KYARK*KL*KSGCK*FG*NCS*RKHGFR*YPC	TNCNTLVIG*CKCLGSLV								
Sample_35	HLNMPESSENYRRVVVNNLDKTAVNGNMALDDTHAQIVTP	PWSLVDANAWGV								

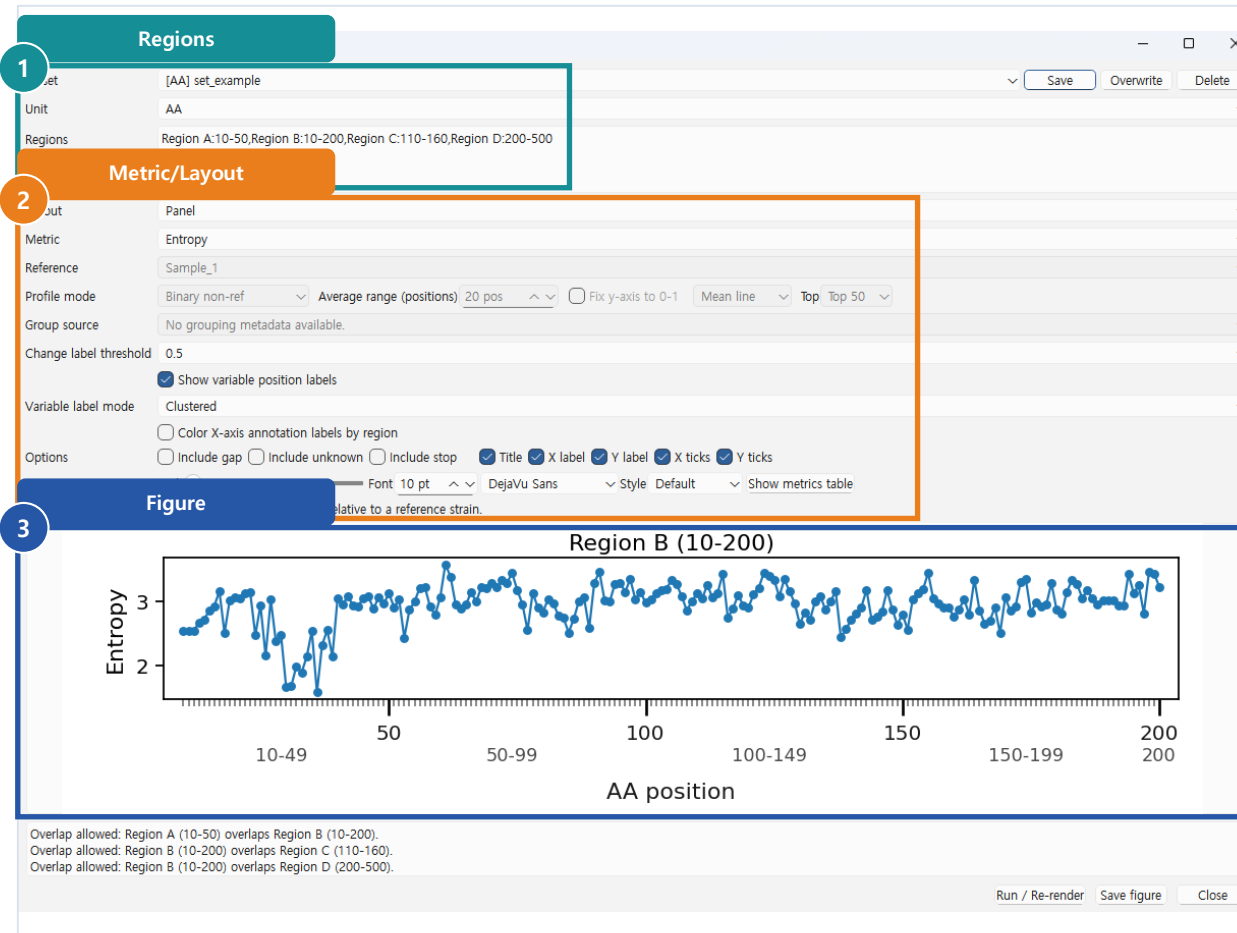
Result Tables

The Counts tab provides position-level summaries, while the Detail tab provides strain- and mutation-level data. Search and sorting help narrow down items quickly.

Key Linkage

Selected mutation information in the table highlights the corresponding strain and position in the viewer, reducing back-and-forth between tables and alignment view.

Region Visualization: Regions and Figures



Region Definition

Define multiple regions using presets or manual input. Layout options include Panel and Concatenate modes.

Metric

Calculate region-centered metrics such as entropy, gap/unknown rate, non-reference rate, mutation map, and strain-wise profiles.

Performance Note

Performing complex region analysis or generating large detail tables for a large number of sequences may be time-consuming.

Region Visualization: Metrics Table

Controls

Table view: Entropy | Sort column: position | Direction: ascending | Scope: within each region

Metrics table

All regions | Region A | Region B | Region C | Region D

	region_label	unit	position	entropy	observed_tokens	token_summary	valid_count	excluded_count
0	Region A	aa	10	2.5299789188029957	A,G,L,P,Q,R,T,V,W	A:1; G:14; L:1; P:2; Q:2; R:4; T:2; V:7; W:1	34	1
1	Region A	aa	11	2.5327195351219203	C,D,G,P,R,S,T,V,W	C:1; D:2; G:13; P:1; R:3; S:3; T:1; V:5; W:1	30	5
2	Region A	aa	12	2.5308006218550014	A,C,G,L,Q,S,T,V,W	A:1; C:1; G:3; L:2; Q:13; S:6; T:1; V:1; W:3	31	4
3	Region A	aa	13	2.6652399286796387	A,C,G,L,M,P,Q,R,S,V	A:1; C:1; G:2; L:7; M:1; P:13; Q:2; R:1; S:3; V:2	33	2
4	Region A	aa	14	2.70768021233142	A,C,G,K,L,Q,R,S,V,W	A:14; C:2; G:1; K:2; L:5; Q:2; R:2; S:4; V:1; W:1	34	1
5	Region A	aa	15	2.8501169519558163	A,C,E,L,N,P,Q,R,S,V,W	A:1; C:5; E:3; L:1; N:1; P:2; Q:1; R:1; S:7; V:11; W:1	34	1
6	Region A	aa	16	2.9220024616785656	A,C,E,K,L,M,N,Q,R,S	A:3; C:4; E:7; K:2; L:2; M:2; N:1; Q:1; R:10; S:2	34	1
7	Region A	aa	17	3.154559938249354	C,E,I,K,M,N,Q,R,S,T,V	C:2; E:1; I:2; K:4; M:5; N:8; Q:4; R:1; S:3; T:1; V:2	33	2
8	Region A	aa	18	2.496877994577654	A,C,D,E,K,L,M,Q,R,Y	A:1; C:2; D:1; E:13; K:10; L:1; M:1; Q:1; R:3; Y:1	34	1
9	Region A	aa	19	3.0130059710789405	A,C,E,K,L,M,N,Q,R,S,T	A:1; C:1; E:5; K:4; L:3; M:2; N:2; Q:1; R:10; S:2; T:1	32	3
10	Region A	aa	20	3.0520121205143798	A,E,G,I,K,L,Q,R,S,T,Y	A:9; E:4; G:2; I:1; K:5; L:5; Q:2; R:1; S:1; T:1; Y:2	33	2
11	Region A	aa	21	3.0487949406953985	A,D,E,G,K,L,Q,R,S,T,W,Y	A:1; D:1; E:1; G:1; K:2; L:2; Q:5; R:5; S:2; T:10; W:1; Y:1	32	3
12	Region A	aa	22	3.124226763509437	A,D,E,G,I,K,L,Q,R,S,T,Y	A:3; D:4; E:1; G:10; I:3; K:1; L:2; Q:2; R:1; S:2; T:1; Y:1	31	4
13	Region A	aa	23	3.136324711025711	D,G,I,K,L,N,Q,R,S,T,W,Y	D:2; G:3; I:1; K:1; L:5; N:1; Q:1; R:2; S:10; T:4; W:3; Y:1	34	1
14	Region A	aa	24	2.468090781663233	D,E,G,I,L,N,S,T,V,W,Y	D:1; E:4; G:18; I:1; L:2; N:1; S:2; T:2; V:1; W:1; Y:1	34	1
15	Region A	aa	25	2.9357417972482365	A,E,G,L,N,R,S,T,W,Y	A:1; E:1; G:6; L:1; N:9; R:3; S:5; T:4; W:3; Y:1	34	1
16	Region A	aa	26	2.147805873542275	E,G,I,L,N,R,S,T,V	E:1; G:20; I:1; L:2; N:1; R:3; S:1; T:2; V:3	34	

Export

Export CSV Close

Table View

Show metrics table opens computed results as tables, including all-region views, per-region tabs, and long-form detail views.

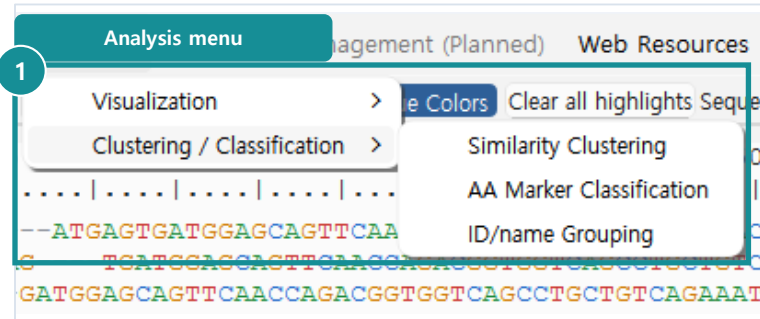
Sorting and Search

Use Table view, Sort column, Direction, Scope, and Search options to narrow down the rows you need.

Export

Use CSV Export to save the current table. For very large detail tables, the preview and full export range may differ.

Grouping and Classification Overview



Similarity Clustering

Groups strains by NT or AA sequence similarity and generates result tables and grouped FASTA outputs.

AA Marker Classification

Defines groups using combinations of conditions at specific AA positions. Example: position 10 = A, position 25 = G.

ID/name Grouping

Classifies sequences based on keywords in the sequence ID or description. Examples: Group_A, Year_2024.

Common Behavior

After successful classification, the setup window closes and the result table is brought forward. If validation or runtime errors occur, the setup window remains open for correction.

Each classification feature uses the current viewer working state and can export grouped FASTA files when needed.

Similarity Clustering

Options

1

Comparison settings

Comparison mode:

☒ NT (nucleotide sequence)
 ☐ AA (amino-acid sequence)

Cluster threshold:

99.00 %

Comparison scope:

Full sequence

100-500

Range is applied to the current comparison mode coordinates.

☐ Include gaps (" ") in comparison (- vs - counts as a match)

NT ambiguity handling (N/R/Y etc.):

Ignore ambiguous positions

AA ambiguity handling (X/*):

Ignore X/*

Result display

☒ Show variation range summary table and highlight on click
 ☐ Show similarity heatmap
 ☒ Reorder heatmap by cluster order
 ☐ Show dendrogram (average linkage)
 ☒ Auto-switch display mode when needed

Export

☐ Save cluster FASTA files
 ☐ Export result bundle to a folder (CSV/FASTA, etc.)
 ☐ Save similarity matrix CSV (large-output caution)

3

Run

Run

Cancel

Options

Choose NT/AA mode, similarity threshold, gap/unknown handling, and comparison range.

Results

Review cluster summaries, pairwise similarity, differing regions, heatmaps, and FASTA bundle export options.

2

Result table

	cluster	size	representative	min_within_similarity(%)	mean_within_similarity(%)	
0	1	5	Sample_16	99.142	99.325	9
1	2	2	Sample_9	99.485	99.485	9
2	3	1	Sample_1	100.0	100.0	1
3	4	1	Sample_2	100.0	100.0	1
4	5	1	Sample_3	100.0	100.0	1
5	6	1	Sample_4	100.0	100.0	1
6	7	1	Sample_5	100.0	100.0	1
7	8	1	Sample_7	100.0	100.0	1
8	9	1	Sample_8	100.0	100.0	1
9	10	1	Sample_10	100.0	100.0	1
10	11	1	Sample_11	100.0	100.0	1
11	12	1	Sample_13	100.0	100.0	1
12	13	1	Sample_14	100.0	100.0	1
13	14	1	Sample_17	100.0	100.0	1
14	15	1	Sample_18	100.0	100.0	1
15	16	1	Sample_19	100.0	100.0	1
16	17	1	Sample_20	100.0	100.0	1

CSV로 내보내기

Cluster Difference Summary (representative = cluster 1)

	ref_cluster	cluster	size	mean_to_ref(%)	diff_sites
0	1	1	5	100.0	0
1	1	2	2	30.092	1224
2	1	3	1	27.535	1271
3	1	4	1	26.319	1290
4	1	5	1	96.73	60
5	1	6	1	30.699	1207
6	1	7	1	29.246	1241
7	1	8	1	29.358	1231
8	1	9	1	29.521	1238
9	1	10	1	30.602	1211
10	1	11	1	29.793	1224
11	1	12	1	26.277	1289
12	1	13	1	29.679	1232
13	1	14	1	25.992	1289
14	1	15	1	26.705	1276
15	1	16	1	28.662	1244
16	1	17	1	27.44	1264

CSV로 내보내기

AA Marker Classification

The screenshot shows the 'Ruleset' window for 'example_set [user]'. It features a 'Rules' tab (orange) and a 'Conditions' tab (blue). The 'Rules' tab contains a list of rules (A, B, C) and buttons for '+ Group', '- Group', 'Copy', and 'Import rules'. The 'Conditions' tab contains a table with columns 'Position' and 'Allowed AA (e.g. D / DE / D,E / * / -X)'. The table has two rows: '1 97 M' and '2 47 A'. Below the table are buttons for '+ Condition' and '- Condition', and a 'Selected group name: A' field. At the bottom, there is a 'Classification options' section with a 'Filter' field (set to 'ID/header filter (substring)'), a 'Stop codon handling' dropdown (set to 'Truncate at first stop'), and two checkboxes: 'Trim NT length to a multiple of 3' (checked) and 'Show/save all matching rules' (unchecked). A 'Run' button is located at the bottom right, with a 'Run' and 'Cancel' sub-button set below it. Numbered callouts are: 1 (Ruleset title bar), 2 (Rules tab), 3 (Conditions tab), and 4 (Run button).

Ruleset

A saved set of marker rules. User-defined rulesets can be saved, copied, and deleted.

Rules and Conditions

Each rule represents one group. Multiple conditions define allowed characters at specific AA positions.

Input Validation

Position accepts digits only. Allowed AA accepts valid amino-acid characters only. Incomplete conditions are not applied.

ID/name Grouping

The screenshot displays the 'Ruleset' window in JU SeqWorkbench Alpha. It is divided into three main sections: 'Ruleset', 'Groups', and 'Patterns'. The 'Ruleset' section at the top shows a dropdown menu set to 'example_set_country' and buttons for 'New', 'Save', 'Delete (user only)', 'Copy...', and 'Rename...'. Below this, the 'Default group (unmatched):' is set to 'Unclassified'. The 'Groups' section on the left lists 'Korea' and 'China' with buttons for '+ Group', '- Delete', and 'Copy'. The 'Patterns' section on the right is titled 'Patterns (one per line, substring matching):' and contains the text 'Korea', 'KR', and 'KOR'. At the bottom, the 'Run/Export' section is open, showing 'Grouping options' with checkboxes for 'Case-sensitive matching' (checked), 'Show result table' (unchecked), 'Save grouped FASTA files' (checked), and 'Show/save all overlapping rules (multi-match)' (unchecked). The 'Matched display' dropdown is set to 'group:pattern'. Below this are checkboxes for 'Remove duplicate Matched values' (checked), 'Prevent overmatching short abbreviations (1-3 characters)' (checked), and 'Match every pattern as a whole token' (unchecked). At the bottom of the 'Run/Export' section are input fields for 'Include' (containing 'Include filter (substring)') and 'Exclude' (containing 'Exclude filter (substring)'), along with 'Run' and 'Cancel' buttons.

1 Ruleset: example_set_country

2 Groups: Korea, China

3 Patterns (one per line, substring matching): Korea, KR, KOR

4 Run/Export: Grouping options

- ☒ Case-sensitive matching
- ☐ Show result table
- ☒ Save grouped FASTA files
- ☐ Show/save all overlapping rules (multi-match)

Matched display: group:pattern

- ☒ Remove duplicate Matched values
- ☒ Prevent overmatching short abbreviations (1-3 characters)
- ☐ Match every pattern as a whole token

Include: Include filter (substring)

Exclude: Exclude filter (substring)

Run Cancel

Keyword-based Grouping

Assign groups using words or patterns contained in the ID or description. Enter one pattern per line.

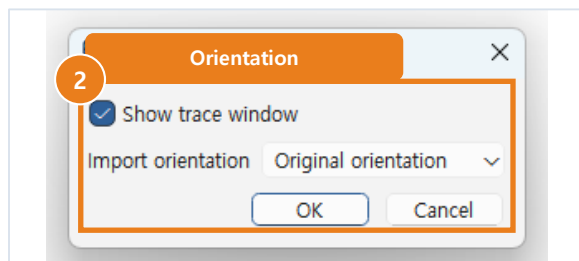
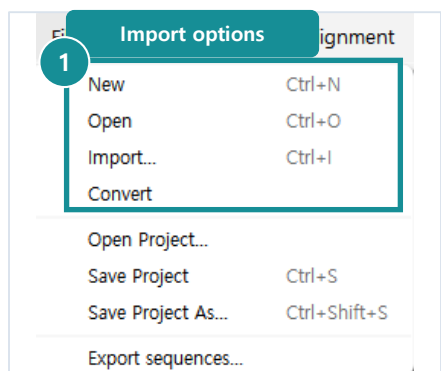
Matching Options

Choose case sensitivity, conservative boundary matching, first/all match behavior, and matched-display style.

Export

Supports classification result tables and grouped FASTA export. AA-only records are written to separate *_aa_only FASTA files.

Sanger AB1 Chromatogram Workflow



Trace Preview

Open AB1/ABI files to inspect chromatograms, basecalled sequences, and quality summaries.



Import

Select Original or Reverse complement orientation, adjust the trim range if needed, and import the sequence into the viewer.

Alpha Notes

Current Focus

This alpha focuses on Sanger/FASTA/MSA sequence review, editing, grouping, and point/region visualization.

Feedback

Please report bugs, documentation issues, and required output formats through GitHub Issues/Discussions or the developer contact.

Out of Scope

It is not a replacement for NGS mapping, variant-calling pipelines, genome assembly, or clinical diagnostic systems.

Performance Limits

Large alignments, complex analyses, and large region detail tables may take time to process. Save your project before running large operations.

Alpha Notes

Alpha Notice and Disclaimer

JU SeqWorkbench Alpha is a pre-release build intended for testing and research workflow validation. The completeness and accuracy of its functions and analytical results are not guaranteed. Users should independently verify important research results and maintain backup copies of important FASTA, AB1/ABI, and project files.

Data and Privacy

JU SeqWorkbench Alpha is a local desktop application. Unless users independently upload or share data outside the application, sequence and project data are processed locally on the user's computer and are not transmitted to external servers.

External Tools

External alignment tools such as MAFFT are not bundled with JU SeqWorkbench Alpha and must be installed separately by the user. Their behavior and results may vary depending on the third-party software implementation, version, and configuration. Important results should therefore be independently verified.